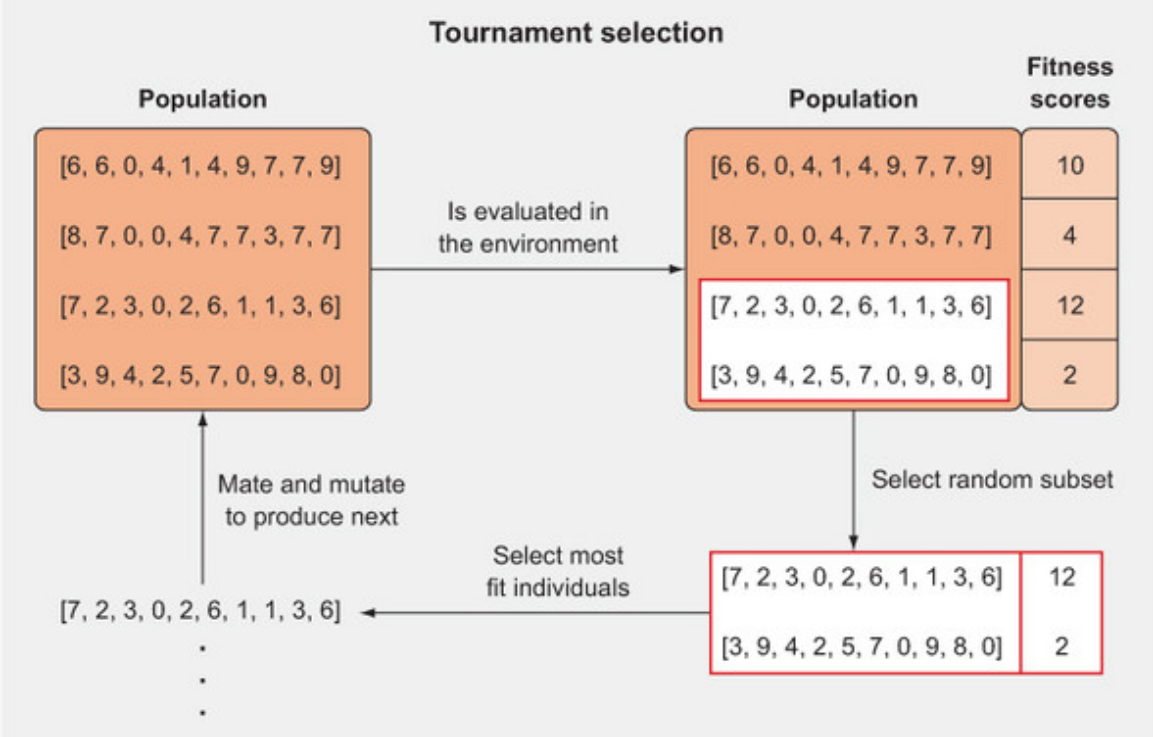


In tournament-style selection we select a random subset from the whole population and then choose the top two individuals in this subset as the parents. This ensures we don't always select the same top two parents, but we do end up selecting the better-performing agents more often.

We can change the *tournament size* (the size of the random subset) to control the degree to which we favor choosing the best agents in the current generation, at the risk of losing genetic diversity. In the extreme case, we could set the tournament size to be equal to the size of the population, in which case we would only select the top two individuals in the population. At the other extreme, we could make the tournament size 2, so that we are randomly selecting parents.

In this example we set the tournament size as a percentage of the size of the population. Empirically, tournament sizes of about 20% seem to work fairly well.

Listing 6.12. Creating the next generation

```
def next_generation(pop,mut_rate=0.001,tournament_size=0.2):
    new_pop = []
```